A Tight Contracting Diffusion Whose Laws Do Not Converge

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Status. Full counterexample, likely valid, pending human review. The time-dependent curvature and integrability hypotheses in the source do not force weak convergence, even for a smooth one-dimensional diffusion with constant nondegenerate noise.

1 Question and counterexample

For a time-inhomogeneous diffusion

$$dX_t = \sigma(t, X_t)\, dB_t + b(t, X_t)\, dt,$$

Cattiaux and Guillin assume

$$|\sigma(t,x) - \sigma(t,y)|^2_{\mathrm{HS}} + 2\langle b(t,x) - b(t,y), x - y\rangle \le -K'(t)|x-y|^2. \tag{1}$$

In Remark 4.5 they observe that if $K(t) \to +\infty$ and $\int_0^\infty e^{-K(s)} ds < \infty$, the transition laws $P(t, x, \cdot)$ are tight, but state that they do not know whether this family converges weakly [1, p. 29]:

> **Remark 4.5.** Assume that $K(t) \to +\infty$ as $t \to +\infty$ and that $C = \int_0^{+\infty} e^{-K(s)}\, ds < +\infty$. Then, for all t, $P(t, x, .)$ (the distribution of the process starting from x and $t_0 = 0$) satisfies a Poincaré inequality (and a log-Sobolev inequality when $\sigma = Id$) with a constant bounded by MC (or $2C$). The family $(P(t, x, .))_{t>0}$ is then tight, but we do not know whether it is weakly convergent or not. Nevertheless any weak limit satisfies the same functional inequality.

Figure 1: The convergence question in Remark 4.5 of [1, p. 29].

Theorem 1. *The stated hypotheses do not imply weak convergence. In dimension one, the smooth diffusion*

$$dX_t = dB_t - \frac{1}{2} X_t\, dt + \sin(t)\, dt, \qquad X_0 = x, \tag{2}$$

satisfies (1) with $K(t) = t$ and has a tight, nonconvergent family of laws. More precisely,

$$\mathcal{L}(X_{2\pi n}) \Longrightarrow \mathcal{N}(-4/5, 1), \qquad \mathcal{L}(X_{(2n+1)\pi}) \Longrightarrow \mathcal{N}(4/5, 1).$$

2 Verification

Here $\sigma \equiv 1$ and $b(t, u) = -u/2 + \sin t$. Therefore, for every $u, v \in \mathbb{R}$,

$$|\sigma(t,u) - \sigma(t,v)|^2 + 2(b(t,u) - b(t,v))(u - v) = -|u - v|^2.$$

Thus (1) holds with equality for $K'(t) = 1$, so one may take $K(t) = t$. Both remaining assumptions are immediate:

$$K(t) \longrightarrow +\infty, \qquad \int_0^\infty e^{-K(s)}\, ds = \int_0^\infty e^{-s}\, ds = 1.$$

The coefficients are globally Lipschitz and smooth, so all standard regularity and nonexplosion requirements in the source are satisfied.

Solving (2) by variation of constants gives

$$X_t = e^{-t/2}x + \int_0^t e^{-(t-s)/2} \sin s\, ds + \int_0^t e^{-(t-s)/2}\, dB_s.$$

Consequently X_t is Gaussian with variance

$$v(t) = \int_0^t e^{-(t-s)} ds = 1 - e^{-t} \longrightarrow 1 \tag{3}$$

and mean

$$\begin{aligned} m_x(t) &= e^{-t/2}x + \int_0^t e^{-(t-s)/2} \sin s\, ds \\ &= \frac{2}{5}\sin t - \frac{4}{5}\cos t + \left(x + \frac{4}{5}\right) e^{-t/2}. \end{aligned} \tag{4}$$

The means are uniformly bounded and the variances are at most one; hence the family $\{\mathcal{L}(X_t) : t \geq 0\}$ is tight. But

$$m_x(2\pi n) \longrightarrow -\frac{4}{5}, \qquad m_x((2n+1)\pi) \longrightarrow \frac{4}{5}.$$

Combining these limits with (3) gives the two distinct Gaussian subsequential limits in Theorem 1. The full family therefore cannot converge weakly. □

3 Mechanism and scope

The obstruction is structural: the curvature condition (1) depends only on drift differences and is completely insensitive to an additive forcing $a(t)$. Strict contraction erases the initial condition but does not erase a persistent periodic motion of the common center. Any claim of convergence under this framework therefore needs an additional condition controlling the absolute drift, for example convergence of the forcing or of an instantaneous equilibrium center.

The counterexample answers exactly the universal question in Remark 4.5; it does not contradict the source's functional inequalities, all of which hold uniformly for the translated Gaussian laws above. The calculation is exact and has no computational dependency. The run indexes and exact arXiv id and question wording were checked; no pre-existing packet or duplicate result was found. Because the mechanism is elementary, an uncatalogued prior example is possible even though no direct later answer was located in the bounded audit.

References

[1] P. Cattiaux and A. Guillin, *Semi Log-Concave Markov Diffusions*, arXiv:1303.6884 (2013).